Subfield descent and the higher-rank Suzuki constructions

Human Compression project

Abstract

We account for the mathematical claims in the fixed twenty-page edition of *The residually weakly primitive pre-geometries of the Suzuki simple groups*. Its exhaustive theorem is false. We retain the earlier counterexample and give proofs or explicit corrections for its remaining claims. The surviving construction families follow from subfield descent, cyclic coset coordinates and scalar irreducibility. Several other prescribed intersection lattices are impossible, and a scalar codimension test rules out the mixed construction. This is a complete treatment of the fixed source's assertions, not a corrected classification of all Suzuki geometries.

1 The source and the meaning of completion

The source is D. Leemans, Note di Matematica 20(1) (2000/2001), 1–20, received 26 July 1999 and accepted 21 September 1999. We use the retained published edition, whose physical and printed page numbers agree. The exact source file and page references are in the claim ledger. The companion volume includes all the proof texts used below, and its acquisition record binds their earlier independent reviews.

For $G_i \leq G$ and a finite type set I, write $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The source's condition is

$$\text{for every } J \subsetneq I \text{ there is } i \notin J \text{ with } G_{J\cup\{i\}} \text{ maximal proper in } G_J. \tag{1}$$

The source uses left cosets and left translation. Its residues for flags inside the distinguished chamber are the algebraic intersection systems. Its local two-transitivity condition applies to those algebraic rank-one residues. Actual thinness counts all chambers incident with an actual panel. The distinction is necessary for the corrections imported from the rank-three treatment. On p. 5 the source defines groups of Suzuki type to be simple and explicitly excludes Sz(2). We apply that convention to the Suzuki labels in its residue lists. In particular the common Suzuki bottom in R_{10}, R_{11}, R_{13} is simple; the parameter is at least eight.

We use flag transitivity in the source's stated strong sense, including transitivity on every flag type. In the positive results here we prove that every flag has a common representative in G, which implies both flag transitivity and the geometry axiom. No inference from chamber transitivity alone is needed.

The source explicitly states on p. 3 that its principal theorem concerns pregeometries, without assuming flag transitivity or the geometry axiom. On p. 15 it requires the *whole intended intersection lattice* of each construction, as inferred from the corresponding rank-four case. A list of abstract parabolic isomorphism types is therefore weaker than the construction being asserted. We keep these two meanings separate. A refuted exhaustive theorem is disposed of by a counterexample; it does not create an obligation to prove a stronger repaired classification.

2 Common theory and the rank-three input

The classical premises are the four-dimensional Suzuki model, its maximal-subgroup and centralizer structure, simple subfield maximality, Lang–Steinberg surjectivity and Schur–Zassenhaus. None is presented as new. The source itself cites Lüneburg for the subgroup

structure. The written common theory in the volume supplies the needed matrix intersections, torus arithmetic, normalizer restrictions and affine uniqueness arguments. The source's own exhaustive rank-three list is never used as a premise.

Lemma 1, that every rank-three coset pregeometry is a geometry, has the short common-representative proof in the acquired preliminary paper. Lemma 2 is the odd-exponent gcd identity for $2^{2m}+1$. Lemma 3 is the proper-overgroup statement for an embedded full split dihedral in a maximal simple subfield; the ambient group itself must be excluded from its literal word "subgroups." Lemma 4 follows from the proved exclusion of the other maximal subgroup types in rank at least four. The consequences of (1) on pp. 4–5, including strict intersections and a maximal chain obtained by ordering labels, are proved in the common prime-degree argument.

Theorems 1 and 2 are recovered: the three necessary prime-degree patterns are proved without an enumeration assumption, each allowed pattern has an explicit realization, and a prime field degree with composite $q-1$ admits no rank at least three satisfying (1). The thirteen labels in Theorem 3 have the following exact relation to the rank-three treatment.

L110 labels	Rank-three source labels	Disposition
R_1, R_2, R_3	Theorem 5.3, patterns 1–3	necessary and realizable
R_4, R_5, R_6	Theorem 5.5, patterns 1–3	prescribed meets impossible
R_7	Theorem 5.5, pattern 4	canonical realization
R_8	Theorem 5.5, pattern 5	prescribed meet impossible
R_9, R_{10}, R_{11}	Theorem 5.5, patterns 6–8	canonical realizations
R_{12}	Theorem 5.5, pattern 9	bare meets exist; (1) fails
R_{13}	Theorem 5.5, pattern 10	canonical realization

The exhaustive assertion in Theorem 3 is false: the acquired $\mathrm{Sz}(8), \mathrm{Sz}(32), C_{533} : C_4$ family in $\mathrm{Sz}(32768)$ has intersection orders $20, 52, 164, 4$, whereas the applicable printed pattern requires the two simple parabolics to meet in order eight. Its geometry, firmness, residual connectedness and flag transitivity are proved. The exact corrections, including the scalar obstruction to R_{12}, are imported with their proofs rather than rederived by a new search.

Theorem 4 is explicitly a quotation of the separate dihedral classification, reference [20]. That original is outside the fixed accessible intake. We retain the quotation as an attributed premise, without claiming recovery of the cited original. The needed odd-dihedral maximality consequences also have elementary proofs in the common theory and the prime-residue extension paper; the higher constructions do not rely on an unchecked machine enumeration of dihedral geometries.

3 The rank-four extension argument

The new nonextension paper proves that none of R_1, R_2, R_3 extends to a rank-four system with its simple parabolic maximal in the ambient Suzuki group. For R_1, the affine subgroup forces its containing soluble parabolic into a point stabilizer, which cannot contain the required odd dihedral. For R_2, the common-reflection argument aligns the two prime-degree simple subfields and makes their meet $\mathrm{Sz}(2)$, of order twenty, not the prescribed dihedral of order ten. For R_3, the common full torus forces an affine meet, and the unique simple subfield containing that embedded affine group makes two parabolics equal. Equal orders alone would not justify that last step. Thus the source's claimed surviving R_2 candidate and Figure 1 are refuted, while the nonextension conclusions for R_1, R_3 are recovered. Section 3.2 follows from the recovered prime-degree obstruction.

In Section 3.3, the R_4, R_5, R_6, R_8 extension branches already start from impossible full residue lattices. Their higher-rank continuations cannot repair the obstruction. The R_{12} branch

fails (1) by the scalar codimension test. The other five branches require a separate necessity argument: existence of the canonical families does not prove that every extension of a prescribed residue has their lattice.

The split extension paper proves this necessity for R_7 and R_9. A field-generating cyclic complement identifies the embedded subfield inside every simple overgroup of the prescribed dihedral or central affine subgroup. Its intersection with the maximal parabolic then excludes such a simple overgroup. The remaining dihedral parabolic lies in the ambient split normalizer; the affine parabolic lies in the appropriate point stabilizer. Scalar irreducibility and the absence of intermediate overgroups force the full central radical and the same cyclic complement in both soluble parabolics. The last cyclic and dihedral maximality conditions force two distinct primes and every remaining intersection. This proves the conditional extension statement also in the repeated R_7 branch of Section 3.3.2. On p. 13 the stray G_4 means G_3. The printed divisibility for the new prime must be $n' \mid (s'-1)/(t-1)$; the quotient $(q-1)/(s'-1)$ is not the correct condition. These textual corrections do not weaken the necessity claim.

For R_{10} and R_{11}, use $H = C_m : C_c$ with the full torus complement, of order two in the split case and four in the nonsplit case. Its normalizer is H itself. Conjugacy inside each subfield group therefore makes the containing subfield of each degree unique. Two such groups containing H meet in the compatible gcd field; in the nonsplit argument the full rotation order exceeds five. We do not assume that the bottom torus of order five determines Sz(8). A divisor argument supplies the remaining information: an intersection system of r cyclic extensions with a common complement has r prime-index steps, and strictness supplies r different prime valuation witnesses. Its top rotation order divided by its bottom order is thus a product of r distinct primes, with one omitted at each parabolic. This remains true when a selected prime divides the bottom order.

The torus necessity paper also derives the input field degrees from (1). A composite relative degree would insert an intermediate compatible torus order, contradicting the prime quotient unless the simple bottom were maximal, which is impossible at that composite degree. Applying the same argument in the ambient residue gives the two distinct relative primes and their product. The source's R_{11} row had not stated those degree identities explicitly. With these premises proved, the fixed maximal simple parabolic determines the other two simple parabolics, and the full-complement and divisor arguments determine the remaining torus parabolic and all its intersections. The R_{10} paragraph's later interchange of G_2, G_3 is corrected to agree with its opening lattice. For R_{13} the common simple bottom and uniqueness align all field groups; maximal prime degrees then give every intersection.

These are conditional necessities for the exact source residues. They do not make the source's global branch list exhaustive. The first assembly conflated these two questions; its independent review held completion on that gap. That assembly and hold are preserved. The two new necessity papers supply the missing arguments in this edition, while the global exhaustive conclusions remain refuted.

4 Every numbered construction

Put $S_d = \mathrm{Sz}(2^d)$. In the positive families choose distinct odd primes a_i, an odd $e > 1$, $t = 2^e$, $n = e\prod_i a_i$, and standard field parabolics S_{n/a_i}. Selecting a set J of field labels leaves S_{n_J} with $n_J = e\prod_{i\notin J} a_i$. The following table states the complete dispositions; the accompanying papers give all subgroup and residue proofs.

No.	Disposition of the source recipe
1	Its intended full lattice contains the forbidden shared-reflection prime-subfield meet. It is impossible; the additional affine parabolic cannot repair it.

No.	Disposition of the source recipe
2–4	The intended lattices contain the impossible R_4, R_5, R_6 residues. Separately, their literal full intersections of orders 10, 4, 4 cannot be subgroups of the displayed odd dihedral: its rotation order has no factor 5 and its two-part is 2. Deleting the last index would repair that notation, but would not repair the intended lattice.
5	After restoring the missing factor $t-1$ in the dihedral order and correcting the conflicting indices and the undefined product index, the canonical split family satisfies (1) for every allowed parameter. The two cases have e prime, including the repeated-prime case $a_i = e$. If several distinguished primes give valid choices, different omitted simple-subfield orders distinguish the resulting stated-action families.
6	The intended R_8 lattice is impossible. This does not exclude the bare parabolic types: the retained example with relative degrees $3, 5, 7$ and torus primes $13, 41, 113$ realizes those types with a different lower lattice, as a thick, residually connected, flag-transitive geometry.
7	Relabel the repeated subfield/dihedral index. The common scalar field lemma proves all intersections and (1) for every allowed parameter.
8–9	Relabel the repeated final subfield/torus index. Every allowed parameter has a canonical realization satisfying (1), thickness, residual connectedness and flag transitivity. Selected primes may divide the base torus order; the valuation proof includes this case.
10	Relabeling the repeated final index does not repair the mathematics. The bare intended intersections exist. For every embedding with the prescribed mixed parabolic and pair intersections, the remaining intersections are all nonmaximal in that parabolic, so (1) fails.
11	Every literal allowed parameter has a canonical thick, residually connected, flag-transitive realization satisfying (1). The exact criterion is $e > 1$, which also permits all distinct prime divisors to be selected when a repeated prime factor remains. That boundary is absent from the source rank bound.

For Constructions 5 and 7 put $M = (t-1)\prod_i \ell_i$, where the ℓ_i are distinct primes dividing $(t^{a_i}-1)/(t-1)$. The two additional parabolics are D_{2M} and $Z_{2^n} : C_M$. The identity

$$\gcd(M, 2^{n_J} - 1) = (t-1)\prod_{i \notin J} \ell_i$$

gives all intersections. That cyclic scalar subgroup generates the full remaining coefficient field, so its affine complement is maximal. The four possibilities for selecting neither, either, or both soluble types then prove (1) for every subset.

For Constructions 8 and 9 the additional group is $K = C_{\gamma\prod_i \ell_i} : C_c$, with $c = 2$ or 4. Compatible tori give $K \cap S_{n_J} = C_{\gamma\prod_{i\notin J}\ell_i} : C_c$. Lang–Steinberg gives a common representative for pairwise incident standard subfield cosets. The Chinese remainder theorem makes $K \longrightarrow \prod_i K/(K \cap S_{n/a_i})$ surjective as a map of coset sets. This proves the common-representative property for the whole family. No nonnormal coset set is treated as a quotient group.

For Construction 10, conjugate the mixed parabolic to $Q = Z_q U_t H_t$. A prescribed pair intersection H surjects onto Q/Z_q, and its overgroups correspond exactly to the $\mathbb{F}_t$-subspaces containing $H \cap Z_q$. Its scalar codimension is $\prod_i a_i - \prod_i a_i/a_j > 1$. Thus every such H has a proper intermediate overgroup in Q. For Construction 11, the field intersection formula and descent give all the positive claims. If $e = 1$, a final $S_1 = 5 : 4$ lies strictly inside a torus normalizer of some S_{a_i}, so (1) fails.

5 The principal correction and final remarks

The earlier accepted principal correction uses

$$n = 3^2 \cdot 5 \cdot 7 \cdot 11 = 3465, \qquad (G_1, G_2, G_3, G_4) = (S_{1155}, S_{693}, S_{495}, S_{315}).$$

Their common intersection is $S_3 = \mathrm{Sz}(8)$ and every available step is a maximal simple-subfield inclusion. All four parabolics are simple, excluding Constructions 1–10, while the literal Construction 11 allows rank at most three for four distinct prime divisors. This refutes Theorem 5. The descent supplement also proves thickness, residual connectedness and flag transitivity. The correction remains an omission from the literal list; it is repaired by replacing that one rank bound with $e > 1$, without thereby obtaining a complete classification.

The source's introductory assertions about local two-transitivity are quotations of the rank-three paper. That paper's complete corrected treatment supplies the exact witnesses: two nonthin rank-three geometries in $\mathrm{Sz}(8)$ and $\mathrm{Sz}(32)$ satisfying the literal algebraic local condition, and a scalar-subspace family of arbitrary rank under the same algebraic convention. The separate new coset-coordinate join now proves that this last family consists of thick, residually connected, flag-transitive geometries with actual local two-transitivity. Thus the quoted rank restriction is false even for actual geometries. Its rank-three members also occur at composite field degrees. The earlier conservative pregeometry proof remains unchanged; this stronger conclusion and its two-page proof are separately identified.

The diagram paper proves the conditional star statements for the surviving prescribed families: the centre is the torus type in 8/9 and the dihedral type in corrected 5/7; pure fields have complete diagrams. The pentagon figure for Construction 1 presupposes an impossible lattice. The future flag-transitivity question is answered for the canonical families 8, 9 and 11 and the retained bare-type example for 6. Flag transitivity of corrected 5/7 and of arbitrary relative positions remains unproved here; it was not an assertion of the source's classification theorem.

The new contribution in this assembly consists of explicit nonextension arguments, the conditional necessity proofs for all viable rank-four branches, the full parameter applications of the common descent and scalar theory, the correction ledger, the diagram proof, and the explicit join establishing actual geometries for the scalar family. Earlier project proofs and finite certificates are retained with attribution. The already checked Lean fragment concerns only the abstract coset-coordinate implication; no Suzuki classification or whole manuscript is claimed formalized. Completing this source adds no new principal-refutation count, no ordinary recovery, no new conjecture count and no compression measurement.